GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER III

Name of Department: - Computer Science and Engineering

1. Subject Code: **PCS 307** Course Title: **OOP WITH C++ LAB**
2. Contact Hours: **L:** 0 **T**: 1 **P**: 2
3. Examination Duration (Hrs): **Theory** **Practical**
4. Relative Weight: **CIE** 50 **MSE** 25 **SEE** 50
5. Credits: 2
6. Semester: III
7. Category of Course: DC
8. Pre-requisite: **PCS 101, PCS 102**

9. **Course Outcome**:**	After completion of the course the students will be able to: CO1: Evaluate the basic difference between object-oriented programming and procedural language and their data types. CO2: Implement the programs using C++ features such as object creation, compile time polymorphism, inheritance, abstraction, encapsulation etc. CO3: Design and solve programs that incorporates the use of object-oriented techniques such as abstract classes, pure virtual functions, and constructors. CO4: Create programs based on the concepts of virtual base classes, virtual functions and STL to solve real time problems

*** Describe the specific knowledge, skills or competencies the students are expected to acquire or demonstrate.*

10. **Details of the Course:**

Sl. No.	List of problems for which student should develop program and execute in the Laboratory	Contact Hours
1.	An electricity board charges the following rates to domestic users to discourage large consumption of energy. For the first 100 units: - 60 P per unit For the next 200 units: -80 P per unit Beyond 300 units: -90 P per unit All users are charged a minimum of Rs 50 if the total amount is more than Rs 300 then an additional surcharge of 15% is added.	

<table>
<tr><td></td><td>Implement a C++ program to read the names of users and number of units consumed and display the charges with names</td><td></td></tr>
<tr><td>2.</td><td>Construct a C++ program that removes a specific character from a given string and return the updated string.
Typical Input: computer science is the future
Typical Output: compuer science is he fuure</td><td></td></tr>
<tr><td>3.</td><td>Implement a C++ program to find the non-repeating characters in string.
Typical Input: graphic era university
Typical Output: c g h n p s t u v y</td><td></td></tr>
<tr><td>4.</td><td>You are given an array of elements. Now you need to choose the best index of this array. An index of the array is called best if the special sum of this index is maximum across the special sum of all the other indices. To calculate the special sum for any index you pick the first element that is and add it to your sum. Now you pick next two elements i.e., and and add both of them to your sum. Now you will pick the next elements, and this continues till the index for which it is possible to pick the elements. Find the best index and in the output print its corresponding special sum. Note that there may be more than one best index, but you need to only print the maximum special sum.

Input
First line contains an integer as input. Next line contains space separated integers denoting the elements of the array
Output
In the output you have to print an integer that denotes the maximum special sum

<table><tr><td colspan="2">Input/Output Format</td></tr><tr><td>Typical Input</td><td>Expected Output</td></tr><tr><td>5
1 3 1 2 5</td><td>8</td></tr><tr><td>10
2 1 3 9 2 4 -10 -9 1 3</td><td>9</td></tr></table></td><td></td></tr>
<tr><td>5.</td><td>Implement a C++ program to demonstrate the concept of data abstraction using the concept of Class and Objects</td><td></td></tr>
<tr><td>6.</td><td>Define a class Hotel in C++ with the following specifications
Private members
• Rno Data member to store room number
• Name Data member to store customer name
• Tariff Data member to store per day charges
• NOD Data member to store number of days of stay
• CALC() Function to calculate and return amount as NOD*Tariff ,and if the value of days* Tariff >10000,</td><td></td></tr>
</table>

	then total amount is 1.05* days*Tariff. **Public members** • Checkin() Function to enter the content Rno, Name, Tariff and NOD • Checkout() Function to display Rno, Name, Tariff, NOD and Amount (amount to be displayed by calling function) CALC()	
7.	Implement a Program in C++ by defining a class to represent a bank account. Include the following: **Data Members** ● Name of the depositor ● Account number ● Type of account (Saving, Current etc.) ● Balance amount in the account **Member Functions** ● To assign initial values ● To deposit an amount ● To withdraw an amount after checking the balance ● To display name and balance	
8.	Anna is a contender for valedictorian of her high school. She wants to know how many students (if any) have scored higher than her in the exams given during this semester. Create a class named Student with the following specifications: ➢ An instance variable named scores holds a student's 5 exam scores. ➢ A void input () function reads 5 integers and saves them to scores. ➢ An int calculateTotalScore() function that returns the sum of the student's scores. **Input Format** In the void Student::input() function, you must read 5 scores from standard input and save them to your scores instance variable. **Output Format** In the int Student::calculateTotalScore() function, you must return the student's total grade (the sum of the values in scores). The code in the editor will determine how many scores are larger than Anna's and print that number to the console. **Sample Input** The first line contains n, the number of students in Anna's class. The n subsequent lines contain each student's 5 exam grades for this semester. 3 30 40 45 10 10	

	40 40 40 10 10 50 20 30 10 10 **Sample Output** 1	
9.	Construct a Program in C++ to show the working of function overloading(compile time polymorphism) by using a function named calculate Area () to calculate area of square, rectangle and triangle using different signatures as required.	
10	Create a class called Invoice that a hardware store might use to represent an invoice for an item sold at the store. An Invoice should include four pieces of information as instance. **Data Members -** • partNumber (type String) • partDescription (type String) • quantity of the item being purchased (type int) • price_per_item (type double) Your class should have a constructor that initializes the four instance variables. Provide a set and a get method for each instance variable. In addition, provide a method named getInvoiceAmount() that calculates the invoice amount (i.e., multiplies the quantity by the price per item), then returns the amount as a double value. If the quantity is not positive, it should be set to 0. If the price per item is not positive,it should be set to0.0. Write a test application named invoiceTest that demonstrates class Invoice's capabilities.	
11	Imagine a tollbooth with a class called TollBooth. The two data items are of type unsigned int and double to hold the total number of cars and total amount of money collected. A constructor initializes both of these data members to 0. A member function called payingCar( )increments the car total and adds 0.5 to the cash total. Another function called nonPayCar( ) increments the car total but adds nothing to the cash total. Finally a member function called display( )shows the two totals. Include a program to test this class. This program should allow the user to push one key to count a paying car and another to count a non paying car. Pushing the ESC key should cause the program to print out the total number of cars and total cash and then exit.	
12	Create a class called Time that has separate int member data for hours, minutes and seconds. One constructor should initialize this data to 0, and another should initialize it to fixed values. A member function should display it in 11:59:59 format. A member function named add() should add two objects of type time passed as arguments. A main ( ) program should create two initialized values together, leaving the result in the third time variable. Finally it should display the value of this third variable.	

13	Create class SavingsAccount. Use a static variable annualInterestRate to store the annual interest rate for all account holders. Each object of the class contains a private instance variable savingsBalance indicating the amount the saver currently has on deposit. Provide method calculateMonthlyInterest() to calculate the monthly interest by multiplying the savingsBalance by annualInterestRate divided by 12.This interest should be added tosavingsBalance. Provide a static method modifyInterestRate() that sets the annualInterestRate to a new value. Write a program to test class SavingsAccount. Instantiate two savingsAccount objects, saver1 and saver2, with balances of Rs2000.00 and Rs3000.00, respectively. Set annualInterestRate to 4%, then calculate the monthly interest and print the new balances for both savers. Then set the annualInterestRate to 5%, calculate the next month's interest and print the new balances for both savers	
14	Create a class Complex having two int type variable named real & img denoting real and imaginary part respectively of a complex number. Overload +, - , == operator to add, to subtract and to compare two complex numbers being denoted by the two complex type objects	
15	Using the concept of operator overloading. Implement a program to overload the following: a. Unary – b. Unary ++ preincrement, postincrement c. Unary -- predecrement, postdecrement	
16	Using the concept of operator overloading. Implement a program to overload the following: With the help of friend function a. Unary – b. Unary ++ preincrement, postincrement c. Unary -- predecrement, postdecrement	
17	Create a Base class that consists of private, protected and public data members and member functions. Try using different access modifiers for inheriting Base class to the Derived class and create a table that summarizes the above three modes (when derived in public, protected and private modes) and shows the access specifier of the members of base class in the Derived class.	
18	You are given three classes A, B and C. All three classes implement their own version of func. In class A, func multiplies the value passed as a parameter by 2. In class B, func multiplies the value passed as a parameter by 3. In class C, func multiplies the value passed as a parameter by 5.You are given class D such that You need to modify the class D and implement the function update_val which sets D's val to new_val by manipulating the value by only calling the func defined in classes A, B and C.It is guaranteed that new_val has only 2, 3 and 5 as its prime factors. Implement class D's function update_val. This function should update D's val only by calling A, B and C's func.	

	Sample Input new_val = 30 **Sample Output** A's func called 1 times B's func called 1 times C's func called 1 times	
19	Create a class called Student that contains the data members like age, name, enroll_no, marks. Create another class called Faculty that contains data members like facultyName, facultyCode, salary,deptt, age, experience, gender. Create the function display() in both the classes to display the respective information. The derived Class Person demonstrates multiple inheritance. The program should be able to call both the base classes and displays their information. Remove the ambiguity (When we have exactly same variables or same methods in both the base classes, which one will becalled?) by proper mechanism	
20	Implement a real case scenario by a proper C++ code to provide the solution to Diamond Problem in C++	
21	Create a base class called shape. Use this class to store two double type values that could be used to compute the area of figures. Derive two specific classes called triangle and rectangle from base shape. Add to the base class, a member function get_data() to initialize base class data members and another member function display_area() to compute and display the area of figures. Make display_area() as a virtual function and redefine this function in the derived class to suit their requirements. Using these three classes, design a program that will accept dimensions of a triangle or a rectangle interactively and display the area. Remember the two values given as input will be treated as lengths of two sides in the case of rectangles and as base and height in the case of triangle and used as follows: Area of rectangle = x * y Area of triangle = ½ *x*y	
21.	Create a base class called CAL_AREA(Abstract). Use this class to store float type values that could be used to compute the volume of figures. Derive two specific classes called cone, hemisphere and cylinder from the base CAL_AREA. Add to the base class, a member function getdata ( ) to initialize base class data members and another member function display volume( ) to compute and display the volume of figures. Make display volume ( ) as a pure virtual function and redefine this function in the derived classes to suit their requirements. Using these three classes, design a program that will accept dimensions of a cone, cylinder and hemisphere interactively and display the volumes. Remember values given as input will be and used as follows:	

	Volume of cone = $(1/3)\pi r^2 h$ Volume of hemisphere = $(2/3)\pi r^3$ Volume of cylinder = $\pi r^2 h$	
22.	The task is to debug the existing code to successfully execute all provided test files. You are required to extend the existing code so that it handles the std::invalid_argument exception properly. More specifically, you have to extend the implementation of the process_input function. It takes integer n as an argument and has to work as follows: 1.It calls function largest_proper_divisor(n). 2.If this call returns a value without raising an exception, it should print in a single line result=d where d is the returned value. 3.Otherwise, if the call raises an invalid_argument exception, it has to print in a single line the string representation of the raised exception, i.e., its message. 4.Finally, no matter if the exception is raised or not, it should print in a single line returning control flow to the caller after any other previously printed output. **Input Format** The input is read by the provided locked code template. In the only line of the input, there is a single integer n, which is going to be the argument passed to function process input. **Output Format** The output should be produced by the function process_input as described in the statement. **Sample Input** 0 **Sample Output** the largest proper divisor is not defined for n=0 returning control flow to the caller Explanation 1 In the first sample, n = 0, so the call largest_proper_divisor(0) raises an exception. In this case, the function process_input prints two lines. In the first of them, it prints the string representation of the raised exception, and in the second line, it prints returning control flow to the caller. **Sample Input** 9 **Sample Output** result=3	
23.	Templates are the foundation of generic programming, which involves writing code in a way that is independent of any particular type. Write a program that can create a list (create a class list) of given type (int, float, char etc.) and perform insertion and deletion on list object.	
24.	Construct a C++ program to demonstrate different methods of List, Vector and Map in STL (Standard Template Library)	

25.	You are provided with a vector of N integers. Then, you are given 2 queries. For the first query, you are provided with 1 integer, which denotes a position in the vector. The value at this position in the vector needs to be erased. The next query consists of 2 integers denoting a range of the positions in the vector. The elements which fall under that range should be removed. The second query is performed on the updated vector which we get after performing the first query. Input Format The first line of the input contains an integer N. The next line contains N space-separated integers (1-based index). The third line contains a single integer x, denoting the position of an element that should be removed from the vector. The fourth line contains two integers a and b denoting the range that should be erased from the vector inclusive of a and exclusive of b. Output Format Print the size of the vector in the first line and the elements of the vector after the two erase operations in the second line separated by space. Sample Input 6 1 4 6 2 8 9 2 2 4 Sample Output 3 1 8 9 Explanation The first query is to erase the 2nd element in the vector, which is 4. Then, a modified vector is {1 6 2 8 9}, we want to remove the range of 2~4, which means the 2nd and 3rd elements should be removed. Then 6 and 2 in the modified vector are removed and we finally get {1 8 9}	
	Total	

11. Suggested Books:

SL. No.	Name of Authors/Books/Publishers	Edition	Year of Publication / Reprint
	Textbooks		
1.	Herbert Schildt,"The Complete Reference C++", 4th Edition, Tata McGraw Hill, 2003.	4th	2003
2.	Balagurusamy." Object Oriented Programming with C++",8th Edition, Tata McGraw Hill,2020	8th	2020
	Reference Books		
1.	Paul Deitel and Harvey Deitel," C++: How to Program",10 Edition, Pearson	10th	2016

12.	**Mode of Evaluation**	Test / Quiz / Assignment / Mid Term Exam / End Term Exam